



Taking to the skies



Montgolfier Hot-air Balloon France 1783

Basket to hold passengers and a fire fueled by straw to create extra-hot air



J.A.C. Charles & the Robert brothers la Charliere France 1783

Balloon made of rubber-coated silk and filled with hydrogen gas

Wings made of linen stretched over bamboo poles and held in place by wires



Jean-Pierre Blanchard's Steerable Balloon France 1784

Paddles designed to steer balloon

Hydrogen was made by pouring **sulphuric acid** onto half a ton of scrap iron!



Cayley Glider UK 1849

Wing made of fabric stretched over hollow canes



Lilienthal's Normal Apparatus Germany 1894

For thousands of years, people have dreamed of flying. However, getting off the ground successfully proved impossible until the invention of lighter-than-air craft, such as balloons and airships, and research into the principles of flight using gliders.

In 1783, following a test flight carrying a sheep, a duck, and a rooster, the **Montgolfier Hot-air Balloon** took off in Paris, France, with two human passengers. Paris was the center of the new balloon age. Just 10 days later the city saw the launch of the first hydrogen-filled balloon, the